

TECHNICAL SPECIFICATIONS

SECTION 26 32 13

ENGINE GENERATORS

Project: Meridian Data Centre Campus, Phase 1
Navi Mumbai, India
Revision: 2024 (December 12, 2024)

PART 1 - GENERAL

1.1 Summary

This section specifies standby diesel generator sets, control panels, local fuel day tanks, and auxiliary systems for critical power backup.

1.2 References

- ISO 8528-5 - Reciprocating internal combustion engine driven AC generator sets
- BS 5514 - Reciprocating internal combustion engines
- IS 10000 - Methods of tests for internal combustion engines
- NFPA 110 - Standard for Emergency and Standby Power Systems

1.3 Submittal Requirements

A. Contractor shall submit the following for Engineer's approval prior to equipment procurement:

1. Product Data: Technical datasheets detailing all electrical and physical parameters, performance curves, and compliance.
2. Shop Drawings: Dimensional layouts showing clearance requirements, cabling paths, and weights.
3. Compliance Matrix: Clause-by-clause confirmation of conformance. Any deviation must be explicitly noted.
4. Factory Test Certificates and Type Test Certifications.

1.4 Quality Assurance

A. Equipment shall be manufactured by a firm specializing in the equipment class with a minimum of twenty (25) years of continuous commercial manufacturing history.

B. As specified in Section 26 05 00 Part 1.4.3.B, all medium voltage switchgear and high-power electrical components must be supplied with certified test certificates from an internationally recognized independent laboratory (ASTA, KEMA, or equivalent) confirming compliance with IEC standards.

PART 2 - PRODUCTS

2.1 Acceptable Manufacturers

A. Subject to compliance with technical requirements, approved manufacturers are limited to the following:

2.2 Performance and Design Requirements

standby_rating: Standby rating must be ≥ 2500 kVA (2000 kWe) at 0.8 PF. Site ratings must be verified at the project design basis ambient of 45°C.

fuel_autonomy: Bulk fuel storage must support 48 hours of continuous operation at full load.

day_tank: Day tanks must support a minimum of 2.0 hours of continuous operation at full load (requires 1040 L).

block_load: Generator must accept 100% block load in a single step per ISO 8528-5 G3 class.

Injected Contract Defects (Internal Quality Control Audit Targets):

- **DEF-005:** Generator room clause Part 3.2.1.F mandates 'adequate ventilation shall be provided in the generator room to prevent heat buildup,' which is an unverifiable qualitative requirement. (Location: 26 32 13 Part 3.2.1.F)
- **DEF-007:** Section 26 32 13 Part 2.3.8.C demands generator day tanks support ≥ 2.0 hours of runtime at full load (1040 L), which directly conflicts with local NBC fire safety codes capping indoor fuel day tanks at 1000 L. (Location: 26 32 13 Part 2.3.8.C vs NBC Part 4)

PART 3 - EXECUTION

3.1 Installation and Examination

- A. Verify that structural concrete floor slab is fully cured. Verify structural load limit of 2,800 kg/m² is not exceeded in UPS or battery rooms.
- B. Maintain a minimum clear work clearance of 1000 mm in front of all enclosures, and 1200 mm above, in accordance with local Indian Electricity Rules and IEC standards.

3.2 Field Quality Control and Site Commissioning

- A. Startup services shall be performed by factory-authorized technicians.
- B. Commissioning tests must be scheduled and carried out in accordance with the project's overall Cx Test Register. Every testing step must map back to a specific specification clause.

startup_tests: Perform initial start and transient load block acceptance testing.

room_ventilation: Verify room ventilation is adequate to prevent engine overheating.